

Image Pair

Texture Sparseness Analysis
 S_d (SIFT density, $w=0.45$)
 S_g (Gradient, $w=0.30$)
 S_c (GLCM contrast, $w=0.25$)
 $S_{\text{pair}} = \max(\text{P90 tiles})$

Lighting Difference Analysis
 Δa (elevation difference, $w=0.4$)
 $\Delta \varphi$ (azimuth difference, $w=0.6$)
 D_{lighting} from SPICE

Routing Decision

$S \leq 0.35$
 $D \leq 0.20$

SIFT/BF or FLANN
Fast classical method

Quality Gate
(Balanced Profile)

Fail

Cascade
Available?

Pass

$Q \geq \text{threshold}$

$S \geq 0.65$
or $D \geq 0.55$

Otherwise

